I8R-JE-IGBJ(a) Clinical Pharmacology Protocol

Protocol I8R-JE-IGBJ
A Phase 3 Study of Nasal Glucagon (LY900018) Compared to Intramuscular Glucagon for Treatment of Insulin-induced Hypoglycemia in Japanese Patients with Diabetes Mellitus

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Glucagon (LY900018)

Eli Lilly Japan K.K
Japan

Clinical Pharmacology Protocol Electronically Signed and Approved by Lilly:
26 October 2017
Amendment (a) Electronically Signed and Approved by Lilly on date provided below.

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1. Protocol Synopsis

Title of Study:
A Phase 3 Study of Nasal Glucagon (LY900018) Compared to Intramuscular Glucagon for Treatment of Insulin-induced Hypoglycemia in Japanese Patients with Diabetes Mellitus

Rationale:
Nasal glucagon (LY900018) is a powder formulation of human glucagon for the rescue treatment of hypoglycemia packaged in a user-friendly, single-use, nasal dosing device that delivers 3 mg glucagon powder for absorption through the nasal mucosa. The glucagon component of LY900018 is a synthetic single-chain, 29-amino-acid polypeptide identical to human glucagon. LY900018 is being developed for the rescue treatment of hypoglycemia.

Currently, glucagon in a liquid form lacks physical and chemical stability; thus, marketed products require reconstitution of glucagon powder before the product can be administered through intramuscular (IM) injection. LY900018 combines stable, synthetic glucagon and a nasal dosing device that for effective use obviates reconstitution, injection, and moreover, patient inhalation. Because of these advantages, LY900018 may offer a significant improvement in the treatment of severe hypoglycemia occurring outside of the hospital setting.

The aim of this study is to compare LY900018 with IM glucagon (IMG), a currently approved product that requires reconstitution prior to administration through IM injection, in Japanese patients with type 1 diabetes mellitus (T1DM) and patients with type 2 diabetes mellitus (T2DM) who achieve treatment success during controlled insulin-induced hypoglycemia. Treatment success is defined as an increase in plasma glucose (PG) to ≥70 mg/dL or an increase of ≥20 mg/dL from the PG nadir within 30 minutes after receiving glucagon, without the patient receiving additional treatments to increase PG. The nadir is defined as the minimum PG measurement at the time of or within 10 minutes following glucagon administration.

Objectives/Endpoints:

Objectives	Endpoints
Primary To demonstrate that 3 mg LY900018 is non-inferior to 1 mg IMG for the proportion of patients achieving treatment success from insulin-induced hypoglycemia using a non-inferiority margin of 10%	The proportion of patients achieving treatment success defined as either an increase in PG to ≥70 mg/dL or an increase of ≥20 mg/dL from nadir within 30 minutes after administration of glucagon. The nadir is defined as the minimum PG value at the time of or within 10 minutes following glucagon administration.
Secondary • To compare the safety and tolerability of 3 mg LY900018 with 1 mg IMG • To characterize the PK profile of 3 mg LY900018 compared to 1 mg IMG • To characterize the PD profile of 3 mg LY900018 compared to 1 mg IMG	• SAE, TEAEs (including gastrointestinal, nasal, and non-nasal AEs), vital signs • PK parameters include AUC, C_{max}, T_{max} • PD parameters include BG_{max} and T_{max}

Abbreviations: AE = adverse event; AUC = area under the concentration versus time curve; BG_{max} = maximal plasma glucose concentration; C_{max} = maximal concentration; IMG = intramuscular glucagon; PD = pharmacodynamics; PG = plasma glucose; PK = pharmacokinetics; SAE = serious adverse event; TEAE = treatment-emergent adverse event; T_{max} = time to maximal concentration.

Summary of Study Design:
Study I8R-JE-IGBJ is a Phase 3, multicenter, randomized, open-label, active comparator, single-dose, 2-treatment, 2-period crossover study in Japanese patients with T1DM and patients with T2DM.

Treatment Arms and Planned Duration for an Individual Patient:
LY900018: Single 3-mg dose; nasal administration by study site personnel
IMG (GlucaGen, Novo Nordisk A/S): Single 1-mg dose; IM injection by study site personnel
Patients will undergo a screening examination within 28 days prior to enrollment. Patients will be administered a single dose in Periods 1 and 2, which will be separated by a washout period of 3 to 14 days. Patients will return for a follow-up visit 26 to 30 days after the last study treatment.

Number of Patients:
Seventy five patients may be enrolled in order to have at least 66 patients (at least 30 patients with T1DM and T2DM, respectively) complete both periods with evaluable primary outcome. If patients discontinue from the study before completion of both periods with evaluable primary outcome for any reason, the patient may be replaced. Replacement should not occur beyond 75 patients enrolled, if it is expected to have at least 66 patients complete the study.

Statistical Analysis:
A total of 66 completers are required in the study in order to achieve the primary objective with at least 90% power using the following assumptions:

- A treatment success rate of 98% for both treatments
- A non-inferiority margin (NIM) of 10%
- One-sided alpha level of 0.025
- A within-patient correlation of zero between 2 treatment visits

The primary analysis will be a treatment comparison of the percentage of patients, including both T1DM and T2DM, who achieve treatment success. The percentage of patients who achieve treatment success within each treatment group and the difference in the percentages between the 2 treatment groups will be computed. A 2-sided 95% confidence interval (CI) will be obtained from the 1-sample mean of the paired differences in primary outcome (1=outcome observed; 0=outcome not observed) across 2 treatment visits. Non-inferiority of LY900018 will be declared if the upper limit of the 2-sided 95% CI constructed on the difference in percentage (IMG - LY900018) is less than the NIM of 10%.

2. Schedule of Activities

Study Schedule Protocol I8R-JE-IGBJ

	Screening	Period 1		Wash out	Period 2		Follow-up/ED	Additional Follow-up for TE ADA[a]	Comments
Procedure	**Days -28 to -2**	**Day -1**	**Day 1**	**3 to 14 days**	**Day -1**	**Day 1**	**Within 28±2 days after last study treatment**		
Clinical Assessments									
Informed Consent	X								
Randomization			X						
Admission to CRU		X			X				Admission can be rescheduled if PG level is not 90 to 250 mg/dL on Day 1.
Discharge from CRU			X			X			Patient may be discharged 6 hours after glucagon administration. A patient may remain inpatient at the CRU at the discretion of the investigator.
Medical History	X								
Collect Pre-existing Conditions and Adverse Events	X	X	X	X	X	X	X		
Physical Exam	X		240 min			240 min	X		After screening, medical assessment only performed to include medical review and targeted examination, as appropriate.

	Screening	Period 1		Wash out	Period 2		Follow-up/ED	Additional Follow-up for TE ADA[a]	Comments
Procedure	**Days -28 to -2**	**Day -1**	**Day 1**	**3 to 14 days**	**Day -1**	**Day 1**	**Within 28±2 days after last study treatment**		
Height and Weight	X		Pre-hypoglycemia induction			Pre-hypoglycemia induction	X		Height is at screening only. Weight on Day 1 will be collected before the start of the procedure to induce hypoglycemia.
Collect Hypoglycemic Events			X			X			Refer to Section 9.5.5.1.
Collect Concomitant Medications	X	X	X	X	X	X	X		Review patients' insulin regimens to confirm acceptability to undergo the procedure to induce hypoglycemia.
Meal			X			X			After completion of all PK sampling, patients will be provided a carbohydrate-rich meal. The investigator will ensure the patient's PG is stable. A prandial insulin dose will be administered if needed.

	Screening	Period 1		Wash out	Period 2		Follow-up/ED	Additional Follow-up for TE ADA[a]	Comments
Procedure	**Days -28 to -2**	**Day -1**	**Day 1**	**3 to 14 days**	**Day -1**	**Day 1**	**Within 28±2 days after last study treatment**		
Insulin infusion to Induce Hypoglycemia			X			X			Procedures in Period 2 will occur at approximately the same time as those in Period 1. Insulin infusion is stopped once bedside PG is <60 mg/dL.
Bedside PG Monitoring for Safety			X			X			Samples will be collected during the hypoglycemia induction procedure and post study treatment (up to 120 mins). Bedside PG monitoring will be conducted no more than 10 minutes apart while bedside PG is ≥90 mg/dL and no more than 5 minutes apart when bedside PG is <90 mg/dL.
Study Treatment Administration			X			X			Once bedside PG reaches <60 mg/dL, the insulin infusion is stopped and approximately 5 minutes later study drug is administered. Indicates Time = 0 min.

	Screening	Period 1		Wash out	Period 2		Follow-up/ED	Additional Follow-up for TE ADA[a]	Comments
Procedure	**Days -28 to -2**	**Day -1**	**Day 1**	**3 to 14 days**	**Day -1**	**Day 1**	**Within 28±2 days after last study treatment**		
Injection-Site Assessment			Pre-hypoglycemia induction, 90 min			Pre-hypoglycemia induction, 90 min			Injection-site assessment for IMG only.
Nasal Inspection			Pre-hypoglycemia induction, 90 min			Pre-hypoglycemia induction, 90 min			Nasal inspection both for IMG and LY900018.
Vital Signs (Supine Blood Pressure, Pulse Rate, and Body Temperature)	X		Pre-hypoglycemia induction, Predose, 15, 30, 60, 120, 240 min			Pre-hypoglycemia induction, Predose, 15, 30, 60, 120, 240 min	X		Time points may be added for each period if warranted and agreed upon between Lilly and the investigator. Predose time point will be between insulin infusion stop and study treatment.
Single 12-lead ECG (Local)	X						X		
Triplicate 12-lead ECG (Central)			Pre-hypoglycemia induction (30 and 15 min prior to insulin infusion), Predose, 15, 30, 60, 120, 240 min			Pre-hypoglycemia induction (30 and 15 min prior to insulin infusion), Predose, 15, 30, 60, 120, 240 min			Time points may be added for Period 1 and Period 2 if warranted and agreed upon between Lilly and the investigator. Predose time point will be between insulin infusion stop and study treatment.
Laboratory Tests									
Clinical Serology Tests	X								See Appendix 2, Clinical Laboratory Tests, for details.

	Screening	Period 1		Wash out	Period 2		Follow-up/ED	Additional Follow-up for TE ADA[a]	Comments
Procedure	**Days -28 to -2**	**Day -1**	**Day 1**	**3 to 14 days**	**Day -1**	**Day 1**	**Within 28±2 days after last study treatment**		
Clinical Lab Tests (Hematology, Clinical Chemistry, and Urinalysis)	X		Pre-hypoglycemia induction			Pre-hypoglycemia induction	X		See Appendix 2, Clinical Laboratory Tests, for details. Patients do not need to fast for samples at screening or follow-up. At Periods 1 and 2, samples should be collected from patients who have fasted at least 8 hours before any study procedures.
HbA1c	X		Pre-hypoglycemia induction						See Appendix 2, Clinical Laboratory Tests, for details.
Pregnancy Test (Female patients of childbearing potential only)	X	X			X		X		Serum pregnancy test will be performed at screening. Urine pregnancy test will be performed at every admission period and follow-up visit if applicable.
FSH (Female patients only)	X								When needed to confirm postmenopausal status.
Ethanol Testing	X								Additional tests can be done at the discretion of the investigator.

	Screening	Period 1		Wash out	Period 2		Follow-up/ED	Additional Follow-up for TE ADA[a]	Comments
Procedure	**Days -28 to -2**	**Day -1**	**Day 1**	**3 to 14 days**	**Day -1**	**Day 1**	**Within 28±2 days after last study treatment**		
Urine Drug Screen	X								Additional tests can be done at the discretion of the investigator.
PK (Glucagon)			Pre-hypoglycemia induction, Predose, 5, 10, 15, 20, 25, 30, 40, 50, 60, 90, 120, 240 min			Pre-hypoglycemia induction, Predose, 5, 10, 15, 20, 25, 30, 40, 50, 60, 90, 120, 240 min	X	X	Sampling times are relative to the time of study treatment administration (0 min). Predose time point will be between insulin infusion stop and study treatment.
Plasma Glucose for PD			Pre-hypoglycemia induction, -5, Predose, 5, 10, 15, 20, 25, 30, 40, 50, 60, 90, 120, 240 min			Pre-hypoglycemia induction, -5, Predose, 5, 10, 15, 20, 25, 30, 40, 50, 60, 90, 120, 240 min			-5 mins = stop insulin infusion. Sampling times are relative to the time of study treatment administration (0 min). Predose time point will be between insulin infusion stop and study treatment.
Genetic Sample (Stored)		X							Single sample for pharmacogenetic analysis taken prior to/on Period 1 Day -1.

	Screening	Period 1		Wash out	Period 2		Follow-up/ED	Additional Follow-up for TE ADA[a]	Comments
Procedure	**Days -28 to -2**	**Day -1**	**Day 1**	**3 to 14 days**	**Day -1**	**Day 1**	**Within 28±2 days after last study treatment**		
Anti-glucagon Antibodies			Pre-hypoglycemia induction				X	X	In the event of drug hypersensitivity reactions (immediate or non-immediate), samples will be collected as close to event onset as possible, at event resolution, and 30 days following the event. Patients with TE ADA at follow-up/ED will undergo additional follow-up. Refer to Section 9.7.1 for details.
Health Outcome Instruments									
Clarke Hypoglycemia Awareness Survey		X							
Nasal and Non-nasal Score Questionnaire			Pre-hypoglycemia induction, 15, 30, 60, 120 min			Pre-hypoglycemia induction, 15, 30, 60, 120 min			Sampling times are relative to the time of study treatment administration (0 min). Questionnaire will be collected in both LY900018 and IMG treatment groups in both periods.

	Screening	Period 1		Wash out	Period 2		Follow-up/ED	Additional Follow-up for TE ADA[a]	Comments
Procedure	**Days -28 to -2**	**Day -1**	**Day 1**	**3 to 14 days**	**Day -1**	**Day 1**	**Within 28±2 days after last study treatment**		
Edinburgh Hypoglycemia Scale: Experimental Hypoglycemia			Pre-hypoglycemia induction, During hypoglycemia induction, Predose, 15, 30, 60, 120 min			Pre-hypoglycemia induction, During hypoglycemia induction, Predose, 15, 30, 60, 120 min			Sampling times are relative to the time of study treatment administration (0 min). During hypoglycemia induction will be when PG ≤75 mg/dL. Predose time point will be between insulin infusion stop and study treatment.

Abbreviations: CRU = clinical research unit; ECG = electrocardiogram; ED = early discontinuation; FSH = follicle-stimulating hormone; HbA1c = hemoglobin A1c; IMG =intramuscular glucagon; min = minutes; PD = pharmacodynamics; PG = plasma glucose; PK = pharmacokinetics; TE ADA = treatment-emergent antidrug antibodies.

[a] Samples for immunogenicity, should be collected every 12 weeks (84±7 days) until the titer returns to baseline (ie, returns to within a single 2-fold dilution of the baseline titer) or until 1 year after the last dose of study treatment. A time-matched PK (glucagon) sample may be collected at the follow-up immunogenicity assessment(s) if warranted and agreed upon between both the investigator and sponsor.

Note: if multiple procedures take place at the same time point, the following order of the procedure should be used: ECG, vital signs, and venipuncture. Plasma glucose sample should be collected as close as possible of protocol-specified time point.

3. Introduction

3.1. Study Rationale

Nasal glucagon (LY900018) is a powder formulation of human glucagon for the rescue treatment of hypoglycemia packaged in a user-friendly, single-use, nasal dosing device that delivers 3 mg glucagon powder for absorption through the nasal mucosa. The glucagon component of LY900018 is a synthetic single-chain, 29-amino-acid polypeptide identical to human glucagon. LY900018 is being developed for the rescue treatment of hypoglycemia.

Currently, glucagon in a liquid form lacks physical and chemical stability; thus, marketed products require reconstitution of glucagon powder before the product can be administered through intramuscular (IM) injection. LY900018 combines stable, synthetic glucagon and a nasal dosing device that for effective use obviates reconstitution, injection, and moreover, patient inhalation. Because of these advantages, LY900018 may offer a significant improvement in the treatment of severe hypoglycemia occurring outside of the hospital setting.

The aim of this study I8R-JE-IGBJ (IGBJ) is to compare LY900018 with IM glucagon (IMG) with respect to the proportion of patients who achieved treatment success during the controlled insulin-induced hypoglycemia in Japanese patients with type 1 diabetes mellitus (T1DM) and patients with type 2 diabetes mellitus (T2DM). Treatment success is defined as an increase in plasma glucose (PG) concentration to ≥70 mg/dL or an increase of ≥20 mg/dL from the PG nadir within 30 minutes after receiving glucagon, without the patient receiving additional treatments to increase PG. The nadir is defined as the minimum PG concentration at the time of or within 10 minutes following glucagon administration. The design and conduct of Study IGBJ are similar to the completed Phase 3 Study I8R-MC-IGBC (IGBC) (Rickels et al. 2016) and the planned Phase 1 Study I8R-MC-IGBI.

3.2. Background

Hypoglycemia is a common complication in all patients with T1DM and some patients with T2DM who use insulin to reduce blood glucose levels. Use of sulfonylurea and glinide by patients with T2DM may also cause hypoglycemia. Depending on the severity, hypoglycemia causes physical symptoms ranging from weakness, dizziness, and sweating progressing to blurred vision, behavioral changes, progressing to unconsciousness, seizures, and coma, and possibly to death (American Diabetes Association 2017). When emergency services are available in a timely manner, intravenous (IV) glucose supplementation is also an effective treatment. Glucagon for injection is a globally available product currently indicated for the treatment of severe hypoglycemia, and is another important treatment option outside of a clinical setting for people who try to rescue patients with severe hypoglycemia. However, for people without enough medical training, the multi-step reconstitution of glucagon and injection procedure would be complex and daunting with substantial risk of errors (Polonsky et al. 2016). Therefore the needle-free and easy-to-administer formulation of glucagon is desired for patients who have a risk of severe hypoglycemia related to anti-diabetes treatments. LY900018 is a powder formulation of synthetic human glucagon in a user-friendly, single-use, nasal dosing

device which delivers 3 mg glucagon powder. Patients do not need to inhale, as the drug is absorbed from the nasal cavity.

Three clinical trials using LY900018 have been completed in non-Japanese adults with T1DM and T2DM: Study IGBC, Study I8R-MC-IGBA (IGBA), and an actual use study (Study I8R-MC-B002 [B002]). Studies IGBC and IGBA demonstrated comparable safety and efficacy between 3 mg LY900018 and 1 mg injectable glucagon in reversing insulin-induced hypoglycemia in adult patients with T1DM only (Study IGBA) or patients with T2DM and T1DM (IGBC). Study B002 was an actual use study that evaluated the effectiveness of 3 mg LY900018 administered by a trained caregiver to patients with T1DM experiencing moderate to severe hypoglycemia in a real-world environment of work and home. Study B002 demonstrated that 96% of moderate to severe hypoglycemic events were resolved within 30 minutes. Finally, 2 trials were conducted in pediatric patients with T1DM: Study I8R-MC-IGBB (IGBB) and Study I8R-MC-B001 (B001). Studies IGBB and B001 demonstrated effectiveness in rescuing pediatric patients from hypoglycemia.

All adult diabetic patients in the 2 inpatient clinical trials (Studies IGBA and IGBC) underwent hypoglycemia induction through IV insulin under close clinical supervision and were administered either LY900018 or injectable glucagon.

Patients fully recovered from hypoglycemia without additional actions to increase glucose level. Specifically, in Study IGBC, at 30 minutes after glucagon dosing, 98.7% (74 out of 75 patients) of LY900018 and 100% (75 out of 75) of IMG-treated patients achieved treatment success (defined as an increase in PG to $\geq$70 mg/dL or an increase of $\geq$20 mg/dL from nadir). The nadir is defined as the minimum glucose measurement at the time of or within 10 minutes following glucagon administration. The LY900018-treated patient who did not meet the above criteria did achieve both PG $\geq$70 mg/dL and an increase of $\geq$20 mg/dL from nadir at 40 minutes after dosing. Furthermore, the mean time to treatment success in participants with nadir glucose <50 mg/dL were 16 minutes in the LY900018 treatment arm and 13 minutes in the IMG treatment arm, respectively (Rickels et al. 2016).

3.3. Benefit/Risk Assessment

There is no anticipated therapeutic benefit for the patients participating in this study.

This study will expose patients to an insulin-induced hypoglycemia meant to simulate hypoglycemia in a controlled setting. The procedure of insulin-induced hypoglycemia targeting nadir glucose around 50 mg/dL is adopted from a previously completed trial, specifically Study IGBC, which demonstrated the safe use of this method. This is an inpatient procedure in which the patients will be under constant supervision of the clinical research unit (CRU) staff with frequent glucose monitoring. Safety provisions have been considered in that IV glucose will be administered if the patient experiences signs and symptoms of severe hypoglycemia and gauged to require intervention during the experimental procedure, at the discretion of the investigator.

Potential risks associated with LY900018, derived from the known risks of currently existing injected glucagon therapy and from the safety profile observed from clinical trials conducted for LY900018, include nausea, vomiting, allergic reactions, increased blood pressure (BP) and heart rate, headache, dysgeusia, and nasal or ocular events (Glucagon G Novo for Injection Package Insert, 2016; Glucagon G Novo for Injection Interview Form, 2015; GLUCAGON for Injection ITO Package Insert, 2016; GLUCAGON for Injection ITO Interview Form, 2016).

Nausea and Vomiting: Glucagon therapy, including LY900018, may cause nausea and vomiting.

Allergic Reactions: Severe, life-threatening, generalized allergy, including anaphylaxis, can occur with glucagon therapy, including LY900018. Patients who have had hypersensitivity reactions to injectable glucagon, or LY900018 or any of its excipients will be excluded from all clinical studies.

Increased Blood Pressure and Heart Rate: Glucagon exerts positive inotropic and chronotropic effects. A temporary increase in both BP and heart rate may occur after the administration of glucagon, including LY900018.

Headache and Dysgeusia: Use of LY900018 may cause headache and altered taste.

Nasal Symptoms: Use of LY900018 may cause a variety of nasal symptoms, such as runny nose, stuffy nose, nasal discomfort, cough, and sneezing.

Ocular Symptoms: Use of LY900018 may cause a variety of ocular symptoms, such as watery eyes, redness of eyes, and itchy eyes.

No additional potential risks of LY900018 were identified in preclinical safety pharmacology and toxicity studies (Reno et al. 2015).

More information about the known and expected benefits, risks, serious adverse events (SAEs), and reasonably anticipated adverse events (AEs) of LY900018 are to be found in the Investigator's Brochure (IB).

More detailed information about the known and expected benefits and risks of GlucaGen® may be found in the Summary of Product Characteristics (GlucaGen Summary of Product Characteristics, 2015).

4. Objectives and Endpoints

Table IGBJ.1 shows the objectives and endpoints of the study.

Table IGBJ.1. Objectives and Endpoints

Objectives	Endpoints
Primary To demonstrate that 3 mg LY900018 is non-inferior to 1 mg IMG for the proportion of patients achieving treatment success from insulin-induced hypoglycemia using a non-inferiority margin of 10%	The proportion of patients achieving treatment success defined as either an increase in PG to $\geq$70 mg/dL or an increase of $\geq$20 mg/dL from nadir within 30 minutes after administration of glucagon. The nadir is defined as the minimum PG value at the time of or within 10 minutes following glucagon administration.
Secondary • To compare the safety and tolerability of 3 mg LY900018 with 1 mg IMG • To characterize the PK profile of 3 mg LY900018 compared to 1 mg IMG • To characterize the PD profile of 3 mg LY900018 compared to 1 mg IMG	• SAE, TEAEs (including gastrointestinal, nasal, and non-nasal AEs), vital signs • PK parameters include AUC, C_{max}, T_{max} • PD parameters include BG_{max} and T_{max}
Exploratory • Explore the formation of anti-glucagon antibodies to glucagon • To evaluate the recovery from clinical symptoms of hypoglycemia	• Presence of anti-glucagon antibodies • Hypoglycemia symptoms questionnaire

Abbreviations: AE = adverse event; AUC = area under the concentration versus time curve; BG_{max} = maximal plasma glucose concentration; C_{max} = maximal concentration; IMG = intramuscular glucagon; PD = pharmacodynamics; PG = plasma glucose; PK = pharmacokinetics; SAE = serious adverse event; TEAE = treatment-emergent adverse event; T_{max} = time to maximal concentration.

5. Study Design

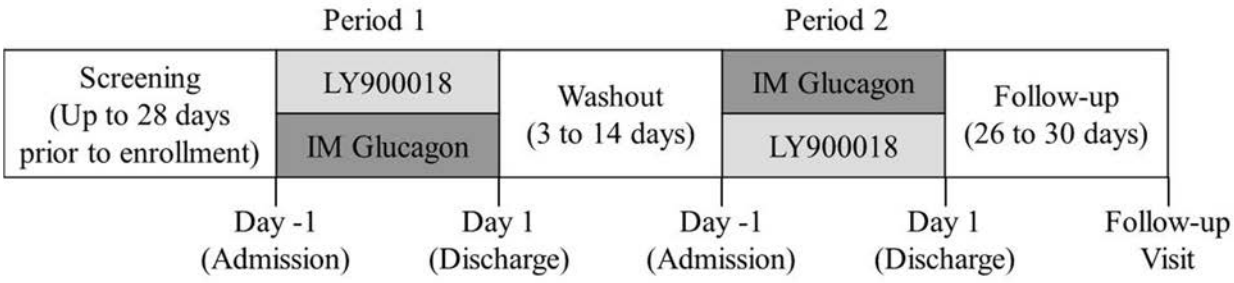
5.1. Overall Design

This is a Phase 3, multicenter, randomized, open-label, active comparator, single-dose, 2-period, 2-treatment, crossover study in Japanese patients with T1DM and T2DM. The study consists of a screening period; treatment period 1 (Period 1); washout period; treatment period 2 (Period 2); follow-up period. Figure IGBJ.1 illustrates the study design. Prior to the study drug administration on Period 1 Day 1, patients will be randomly assigned to a treatment sequence (either LY900018 in Period 1 and IMG in Period 2, or vice versa).

Safety data will be reviewed after the first 6 patients (regardless of type of diabetes) are administered LY900018 in Period 2, and the remaining patients will be dosed after confirmation of the safety. The investigator and Lilly clinical research physician (CRP) or scientist will review available safety data, including AEs, SAEs, vital signs, electrocardiograms (ECGs), and safety laboratory tests, from these patients after they complete Period 2 Day 1. If no clinically significant safety findings for treatment or study procedure are noted, the remaining patients will be dosed.

In each treatment period, patients will undergo a procedure to induce hypoglycemia using IV insulin infusion and serial blood sampling will be conducted to monitor bedside PG for safety. The insulin infusion will be stopped once the PG level reaches <60 mg/dL and approximately 5 minutes later patients will be administered either 3 mg LY900018 or 1 mg IMG (GlucaGen). Serial blood sampling will be performed for glucagon (for pharmacokinetics [PK]) and PG (for pharmacodynamics [PD]) concentration measurements immediately before and up to 4 hours following the administration of glucagon. In each period, patients will remain in the CRU for at least 6 hours after glucagon administration. All patients will receive a carbohydrate-rich meal prior to discharge. Patients may stay longer as needed, at the discretion of the investigator.

Study governance considerations are described in detail in Appendix 3.



Abbreviations: IM = intramuscular.

Figure IGBJ.1. Illustration of study design for Protocol I8R-JE-IGBJ.

5.2. Number of Participants

Seventy five patients may be enrolled in order to have at least 66 patients (at least 30 patients with T1DM and T2DM, respectively) complete the study. For purposes of this study, a completer is defined as a patient who completes both periods with evaluable primary outcome. If patients discontinue from the study before completion of both periods with evaluable primary outcome for any reason, the patient may be replaced to ensure 66 patients complete the study. The replacement patients will be assigned the same treatment sequence as the patients to be replaced and will complete that treatment sequence in its entirety. Replacement should not occur beyond 75 patients enrolled, if it is expected to have at least 66 patients complete the study.

5.3. End of Study Definition

End of the study is the date of the last visit or last scheduled procedure shown in the Schedule of Activities (Section 2) for the last patient.

5.4. Scientific Rationale for Study Design

This study design involves an open-label assessment of 3 mg LY900018 compared to 1 mg of the marketed IMG (GlucaGen). The study will be open label because the different administration routes (ie, intranasal and IM) cannot be blinded. The use of a crossover design allows each patient to serve as his or her own control, thereby reducing variability. The washout period of 3 to 14 days is considered sufficient based on the short half-life of LY900018 ($t_{1/2}$ = approximately 25 minutes). Since the critical role of glucagon in the treatment of severe hypoglycemia is to raise the PG level sufficiently to restore cognition to the point where oral carbohydrate can be ingested, treatment success is defined either as an increase in PG returning to normal level of ≥70 mg/dL or an increase of ≥20 mg/dL from the PG nadir within 30 minutes after receiving glucagon, without the patient receiving additional actions to increase glucose. It is believed that an expected PG nadir of approximately 50 mg/dL (approximately 5 minutes after stop of insulin infusion at 60 mg/dL) is low enough to generate clinical symptoms in most participants and is high enough to avoid impairment of consciousness. The primary analysis includes both patients with T1DM and T2DM since it is expected that the proportion of treatment success is similar between patients with T1DM and T2DM.

5.5. Justification for Dose

Results from studies in both pediatric and adult populations confirm that, while the physiological response to glucagon does appear to saturate between the 2- and 3-mg LY900018 doses, the lower 2-mg dose may not always elicit the maximum response needed in an emergency situation of severe hypoglycemia. The 3-mg dose provides more consistent clinical efficacy compared to the 2-mg dose and is similarly well tolerated in both pediatric and adult patients.

The 1-mg dose of IMG is selected as the approved dose for the rescue treatment of hypoglycemia in Japan.

Based on LY900018 exposure, efficacy, and safety, the 3-mg dose was selected for the Phase 3 studies. Therefore, in the current study, the 3-mg LY900018 dose will be used to compare against 1 mg IMG.

6. Study Population

Eligibility of patients for the study will be based on the results of screening medical history, physical examination, vital signs, clinical laboratory tests, and ECG.

The nature of any conditions present at the time of the physical examination and any preexisting conditions will be documented.

Screening may occur up to 28 days prior to enrollment. Patients who are not enrolled within 28 days of screening may be subjected to an additional medical assessment and/or clinical measurements to confirm their eligibility.

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, are not permitted.

6.1. Inclusion Criteria

Patients are eligible for inclusion in the study only if they meet all of the following criteria at screening and/or enrollment:

[1] have had a diagnosis of either:

[1a] T1DM based on the World Health Organization (WHO) diagnostic criteria, and have been on the following daily insulin therapy for at least 1 year

[A] multiple daily injection of long-acting insulin analog (either insulin glargine [U-100 or U-300] or insulin degludec [U-100]) and rapid-acting insulin analog (insulin lispro, insulin aspart, or insulin glulisine), or

[B] continuous subcutaneous insulin infusion (CSII)

Or

[1b] T2DM based on the WHO diagnostic criteria, and have received the following daily insulin therapy with or without oral anti-hyperglycemic medications (OAMs) for at least 1 year

[A] insulin: long-acting insulin analog (either insulin glargine [U-100 or U-300] or insulin degludec [U-100]) alone, or in combination with rapid-acting insulin analog (insulin lispro, insulin aspart, or insulin glulisine) or CSII

[B] OAM: up to 3 of the following OAMs in accordance with local regulations: metformin, dipeptidyl peptidase-4 inhibitor, sodium glucose cotransporter 2 inhibitor, sulfonylurea (should not be more than half of maximum approved doses), glinides, alpha-glucosidase inhibitor, or thiazolidine

[2] male patients: agree to use an effective method of contraception for the duration of the study and for 28 days following the last study treatment

[3] female patients:

[3a] women of childbearing potential who are abstinent (if this is complete abstinence, as their preferred and usual lifestyle) or in a same sex relationship (as part of their preferred and usual lifestyle) must agree to either remain abstinent or stay in a same sex relationship without sexual relationships with males. Periodic abstinence (eg, calendar, ovulation, symptothermal, post-ovulation methods), declaration of abstinence just for the duration of a trial, and withdrawal are not acceptable methods of contraception.

[3b] otherwise, women of childbearing potential participating must agree to use one highly effective method (less than 1% failure rate) of contraception, or a combination of 2 effective methods of contraception for the entirety of the study.

[A] women of childbearing potential participating must test negative for pregnancy prior to initiation of treatment as indicated by a negative serum pregnancy test at the screening visit followed by a negative urine pregnancy test within 24 hours prior to exposure.

[B] either one highly effective method of contraception or a combination of 2 effective methods of contraception will be used. The patient may choose to use a double barrier method of contraception. Barrier protection methods without concomitant use of a spermicide are not a reliable or acceptable method. Thus, each barrier method must include use of a spermicide. It should be noted that the use of male and female condoms as a double barrier method is not considered acceptable due to the high failure rate when these methods are combined.

[3c] women not of childbearing potential may participate, and include those who are:

[A] infertile due to surgical sterilization (hysterectomy, bilateral oophorectomy, or tubal ligation), congenital anomaly such as mullerian agenesis

Or

[B] postmenopausal, defined as either:

[i] a woman at least 50 years of age with an intact uterus, not on hormone therapy, who has had either:

[a] cessation of menses for at least 1 year, or

[b] at least 6 months of spontaneous amenorrhea with a follicle-stimulating hormone >40 mIU/mL; or

[ii] a woman 55 or older not on hormone therapy, who has had at least 6 months of spontaneous amenorrhea, or

[iii] a woman at least 55 years of age with a diagnosis of menopause prior to starting hormone replacement therapy

[4] are between 18 and 64 years old for T1DM, or between 20 and 70 years old for T2DM at the time of informed consent

[5] have a body mass index of 18.5 to 30.0 kg/m^2 for T1DM, or 18.5 to 35.0 kg/m^2 for T2DM at the time of screening

[6] have a hemoglobin A1c value ≤10% at the time of screening

[7] have clinical laboratory test results within normal reference range (except for glycemic parameters) for the population or investigative site, or results with acceptable deviations that are judged to be not clinically significant by the investigator

[8] have venous access sufficient to allow for blood sampling and administration of insulin for IV administration as per the protocol

[9] are reliable and willing to make themselves available for the duration of the study and are willing to follow study procedures

[10] are able and willing to give signed informed consent

6.2. Exclusion Criteria

Patients will be excluded from study enrollment if they meet any of the following criteria at screening and/or enrollment:

[11] are investigative site personnel directly affiliated with this study and their immediate families. Immediate family is defined as a spouse, biological or legal guardian, child, or sibling

[12] are Lilly employees

[13] are currently enrolled in a clinical trial involving an investigational product or any other type of medical research judged not to be scientifically or medically compatible with this study

[14] have participated, within the last 4 months, in a clinical trial involving an investigational product. If the previous investigational product has a long half-life, 4 months or 5 half-lives (whichever is longer) should have passed from the last dose of investigational product

[15] have previously completed or withdrawn from this study or any other study investigating LY900018, and have previously received LY900018

[16] have known allergies or sensitivity to LY900018, glucagon, related compounds, or any components of the formulation, or history of significant atopy

[17] have an abnormality in the 12-lead ECG that, in the opinion of the investigator, increases the risks associated with participating in the study

[18] any significant changes in insulin regimen and/or unstable blood glucose control within the past 3 months prior to screening as assessed by the investigator

[19] have received a total daily dose of insulin ≥1.2 U/kg at the time of screening

[20] have poorly controlled hypertension (ie, supine systolic BP ≥165 mm Hg or supine diastolic BP ≥95 mm Hg) at screening, or a change in antihypertensive medications within 30 days prior to screening

[21] have a history of pheochromocytoma (ie, adrenal gland tumor) or insulinoma

[22] have a history of an episode of severe hypoglycemia (as defined by an episode that required third party assistance for treatment) in the 1 month prior to screening or have a history of loss of consciousness within the last 2 years induced other than by hypoglycemia

[23] have a history of epilepsy or seizure disorder

[24] have a history or presence of cardiovascular, respiratory, hepatic, renal, gastrointestinal, endocrine (apart from T1DM or T2DM), hematological, or neurological disorders capable of significantly altering the absorption, metabolism, or elimination of drugs; of constituting a risk when taking the investigational product; or of interfering with the interpretation of data

[25] have known or ongoing psychiatric disorders that, in the opinion of the investigator, may preclude the patient from following and completing the protocol

[26] regularly use known drugs of abuse and/or show positive findings on urinary drug screening

[27] show evidence of human immunodeficiency virus (HIV) infection and/or positive HIV antibodies and/or antigen

[28] show evidence of hepatitis C and/or positive hepatitis C antibody

[29] show evidence of hepatitis B and/or positive hepatitis B surface antigen

[30] show evidence of syphilis and/or are positive for syphilis test

[31] are women who are lactating

[32] use of daily systemic beta-blocker, indomethacin, warfarin, anticholinergic drugs

[33] have donated 400 mL or more blood in the last 12 weeks (males) or in the last 16 weeks (females), or any blood donation (including apheresis) within the last 4 weeks, or total volume of blood donation within 12 months is 1200 mL (males)/800 mL (females) or more at screening

[34] have an average weekly alcohol intake that exceeds 21 units per week (males up to age 65) and 14 units per week (males over 65 and females), or are unwilling to stop alcohol consumption from Day -2 to discharge from CRU in each period (1 unit = 12 oz or 360 mL of beer; 5 oz or 150 mL of wine; 1.5 oz or 45 mL of distilled spirits)

[35] in the opinion of the investigator or sponsor, are unsuitable for inclusion in the study

[36] have pre-proliferative and proliferative retinopathy or maculopathy requiring treatment or not clinically stable in the last 6 months, or patients with active changes in subjective eye symptoms as determined by the investigator if an eye exam has not been performed in the last 6 months.
Note: If an eye examination has been performed no more than 6 months before screening, it will not have to be repeated; however, the investigator will need to confirm via interview that there is no change in subjective symptoms.

6.3. Lifestyle and/or Dietary Requirements

Throughout the study, patients may undergo medical assessments and review of compliance with requirements before continuing in the study.

6.3.1. Meals and Dietary Restrictions

In both treatment periods, patients will fast for at least 8 hours prior to beginning the hypoglycemia induction procedure. Patients should not be given any calorie-containing food/drink during insulin-induced hypoglycemia and within 90 minutes of glucagon administration unless there is a safety concern. The patients will receive a carbohydrate-rich meal after completion of all pharmacokinetic (PK) blood sample collections, at 240 minutes post glucagon dose. Patients will be given access to calorie-free water up to 240 minutes post study treatment.

While resident in the CRU, patients should not consume any food or caloric drinks other than that provided by the CRU. When not resident in the CRU, patients will be encouraged to follow their normal diets.

6.3.2. Caffeine, Alcohol, and Tobacco

Patients should refrain from caffeine-containing food/beverages (eg, cola, chocolate drinks, tea, and coffee) from Day -1 to discharge from CRU in each period. Patient alcohol intake should not exceed 21 units per week (males up to age 65) and 14 units per week (males over 65 and females) during the study. Patients must not consume alcohol from Day -2 to discharge from CRU in each period. Smoking will not be permitted when resident in the CRU.

6.3.3. Activity

Patients are encouraged to maintain their regular exercise habits for the duration of the study. However, patients should avoid strenuous exercise 48 hours prior to admission or study visit.

From the beginning of hypoglycemia induction to 240 minutes post study treatment, patients should remain recumbent or sitting on the bed.

6.3.4. Contraception

Male patients must use an effective method of contraception for the duration of the study and for 28 days following the last study treatment.

Female patients of childbearing potential must use one highly effective method of contraception (<1% failure rate; such as combined oral contraceptive pill, implanted contraceptives, or intrauterine device) or a combination of 2 effective methods of contraception (such as male or female condoms with spermicide [not approved in Japan], diaphragms with spermicide [not approved in Japan], or cervical sponges) during the study and for 28 days following the last study treatment. The patient may choose to use a double barrier method of contraception. Barrier protection methods without concomitant use of a spermicide are not a reliable or acceptable method. Thus, each barrier method must include use of a spermicide. It should be noted that the use of male and female condoms as a double barrier method is not considered acceptable due to the high failure rate when these methods are combined.

6.4. Screen Failures

Individuals who do not meet the criteria for participation in this study (screen failure) may not be re-screened.